

Scalability & Future Plan

Date	4 July 2024
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address
1	Supports only structured applicant data.	Limits prediction using additional financial information.	High
2	Runs on a local Flask server.	Cannot be accessed by multiple users over the internet.	High
3	Model requires periodic retraining with new data.	Prediction accuracy may decrease over time.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of local users.	Deploy on a cloud platform with load balancing to support concurrent users.
2	Data Storage	Uses local project files and trained model.	Store datasets and models in cloud storage or a managed database.
3	Performance	Prediction runs on a single local server.	Optimize inference and enable scalable cloud deployment.
4	Security	Basic input validation implemented.	Add user authentication, HTTPS, and role-based access control.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment and public access	1–2 Months	Improve accessibility and support multiple users.
Phase 2	Integration with banking APIs and credit score services	2–3 Months	Enhance prediction accuracy using real-world financial data.
Phase 3	Explainable AI (SHAP/LIME) for prediction transparency	3–4 Months	Help users and banks understand prediction decisions.
Phase 4	Mobile application and real-time analytics dashboard	4–6 Months	Improve user experience and provide monitoring capabilities.